



ST BENEDICT'S

SUBJECT	Chemistry
GRADE	11
EXAMINER	Mr McCoy
MODERATOR	Mr Hilder
DATE	28 November 2022
MARKS	200
DURATION	3 Hours

MARK
TOTAL

LEARNER NAME:

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 22 pages and a Data Sheet of 2 pages. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment. Remember to write your name and surname in the space provided above.
4. Use the data and formulae sheet whenever necessary.
5. Show your working in all calculations.
6. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
7. Answers must be expressed in decimal format, not left as proper fractions.
8. Where appropriate, express answers to TWO decimal places.
9. It is in your own interest to write legibly and to present your work neatly.
10. Blank pages have been included at the end of the paper (pages 22-24). If you run out of space for a question, use this page. Clearly indicate the question number of your answer should you use this extra space.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Question	1	2	3	4	5	6	7	8	9	Total
Mark										
Moderated Mark										
Question Total	20	13	20	15	34	28	23	17	30	200

Comment:

QUESTION 1**20 MARKS**

Answer these questions on the multiple-choice answer sheet below. Make a cross (X) in the box corresponding to the letter that you consider to be correct.

A	B	☒	D
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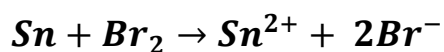
Here the option C has been marked as an example

1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D
1.7	A	B	C	D
1.8	A	B	C	D
1.9	A	B	C	D
1.10	A	B	C	D

1.1 Which of the following is the correct chemical formula for aluminium dichromate? (2)

- A AlCr_2O_7
- B $\text{Al}_2(\text{Cr}_2\text{O}_7)_3$
- C $\text{Al}_3(\text{Cr}_2\text{O}_7)_2$
- D $\text{Al}_3\text{Cr}_2\text{O}_7$

1.2 Consider the net ionic equation:



The chemical species undergoing reduction is... (2)

- A the tin atom.
- B bromine.
- C the tin ion.
- D bromide.

- 1.3 A 5,62 g sample of anhydrous sodium sulfate, Na_2SO_4 , is dissolved in water and the solution is made up to 250 cm^3 in a volumetric flask.
- What is the concentration, in $\text{mol}\cdot\text{dm}^{-3}$, of the hydrated sodium sulfate solution? (2)
- A 0,00016
B 0,0016
C 0,016
D 0,16
- 1.4 Which of the following statements is **NOT** correct? (2)
- A The strongest type of intermolecular force in ammonia is hydrogen bonding.
B The strongest type of intermolecular force in hydrogen chloride is permanent dipole-dipole force.
C The strongest intermolecular force in ethanol is permanent dipole-dipole force.
D The strongest intermolecular force in water is hydrogen bonding.
- 1.5 Which one of the following statements represents the best explanation for the term "heat-of-reaction"? (2)
- A Reactions which transform chemical potential energy into thermal energy.
B The net change of kinetic energy of the system.
C Reactions which transform thermal energy into chemical potential energy
D. The net change of chemical potential energy of the system.
- 1.6 How many **atoms** are present in 1 mol of $(\text{NH}_4)_2\text{SO}_4$ (2)
- A $6,02 \times 10^{23}$
B $6,02 \times 10^{24}$
C $8,43 \times 10^{24}$
D $9,03 \times 10^{24}$
- 1.7 Which of the following is **NOT** true regarding balanced chemical equations? (2)
- A They contain the same number of atoms on each side
B Electrons are also balanced
C An equal number of molecules on both the side
D Follows the law of conservation of mass

1.8 Which ONE of the following statements is CORRECT? (2)

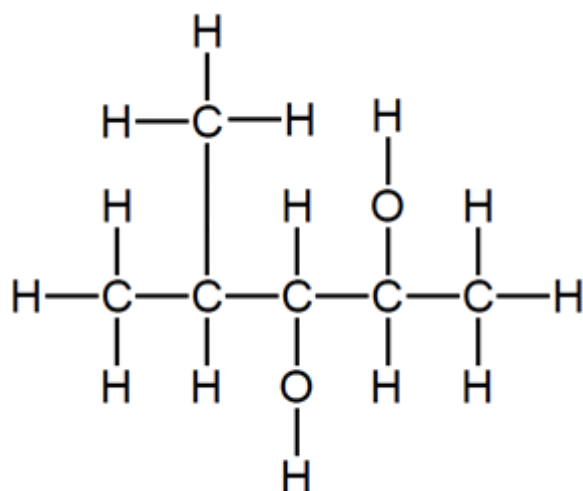
Alkenes ...

- A have the general formula C_nH_{2n+2} .
- B are unsaturated hydrocarbons.
- C readily undergo substitution reactions.
- D have one triple bond between two carbon atoms.

1.9 What is the organic product formed if propene ($CH_3CH=CH_2$) is reacted with steam and a phosphoric acid catalyst? (2)

- A $CH_3CH(OH)CH_2OH$
- B $CH_3CH_2CH_3$
- C $HOCH_2CH_2CH_2OH$
- D $CH_3CH_2CH_2OH$

1.10 What is the IUPAC name of the following compound? (2)



- A 4-methylpenta-2,3-diol
- B 4-methylpentane-2,3-diol
- C 2-methylpenta-3,4-diol
- D 2-methylpentane-3,4-diol

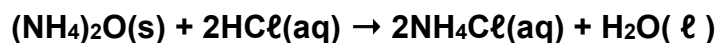
QUESTION 2

13 MARKS

To determine the percentage purity of an impure sample of ammonium oxide, $(\text{NH}_4)_2\text{O}$, a learner follows Steps A and B outlined below:

STEP A:

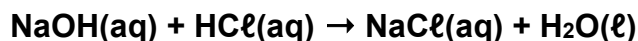
5 g of impure ammonium oxide is dissolved in 100 cm^3 of hydrochloric acid (HCl) of concentration $2\text{ mol}\cdot\text{dm}^{-3}$. The acid (HCl) is added IN EXCESS.



STEP B:

The excess (leftover) HCl is reacted with sodium hydroxide solution (NaOH), to determine how much of the acid was in excess and DID NOT react with the $(\text{NH}_4)_2\text{O}$.

40 cm^3 of $0,6\text{ mol}\cdot\text{dm}^{-3}$ sodium hydroxide solution neutralises the excess hydrochloric acid.



- 2.1 Calculate the number of moles of hydrochloric acid initially added to the ammonium oxide in STEP A. (3)

- 2.2 Calculate the number of moles of excess hydrochloric acid that reacted with the NaOH in STEP B. Answer to 3 decimal places. (3)

- 2.3 Use your answers to 2.1 and 2.2 to calculate how many moles of HCl reacted with the $(\text{NH}_4)_2\text{O}$ in STEP A. Answer to 3 decimal places. (2)

- 2.4 Hence calculate the percentage purity of the original $(\text{NH}_4)_2\text{O}$ sample. Assume that none of the impurities in the $(\text{NH}_4)_2\text{O}$ reacted with the acid. (5)

QUESTION 3**20 MARKS**

- 3.1 The table below gives information about the boiling points of various substances which contain a halogen atom.

Substance	F ₂	Br ₂	HF	HBr	CH ₃ F	CH ₃ Br
Boiling point (°C)	-188	59	20	-66	-78	4

- 3.1.1 Explain why bromine requires more energy to boil than fluorine does. (4)

- 3.1.2 Explain why the boiling points of hydrogen bromide (HBr), and hydrogen fluoride (HF) differ. (5)

3.1.3 The boiling points of fluoromethane (CH_3F) and bromomethane (CH_3Br) differ by approximately 80°C .

Name the strongest type of intermolecular forces in these two substances. (2)

3.2 The melting point of potassium sulfate (K_2SO_4) is 197°C , and that of calcium sulfate is 1460°C .

3.2.1 Write the balanced equation for potassium sulfate dissociating in water. (2)

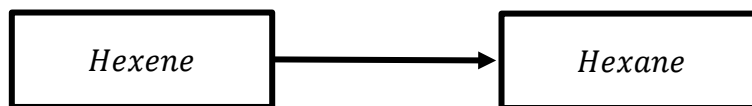
3.2.2 Name the intramolecular bond in potassium sulfate. (2)

3.2.3 Explain the vast difference in melting points of these two substances. (3)

3.2.4 Predict which substance is more soluble in water at room temperature: calcium sulfate or potassium sulfate. (2)

QUESTION 4**15 MARKS**

Study the flowchart below shows the hydrogenation of hexene to hexane.



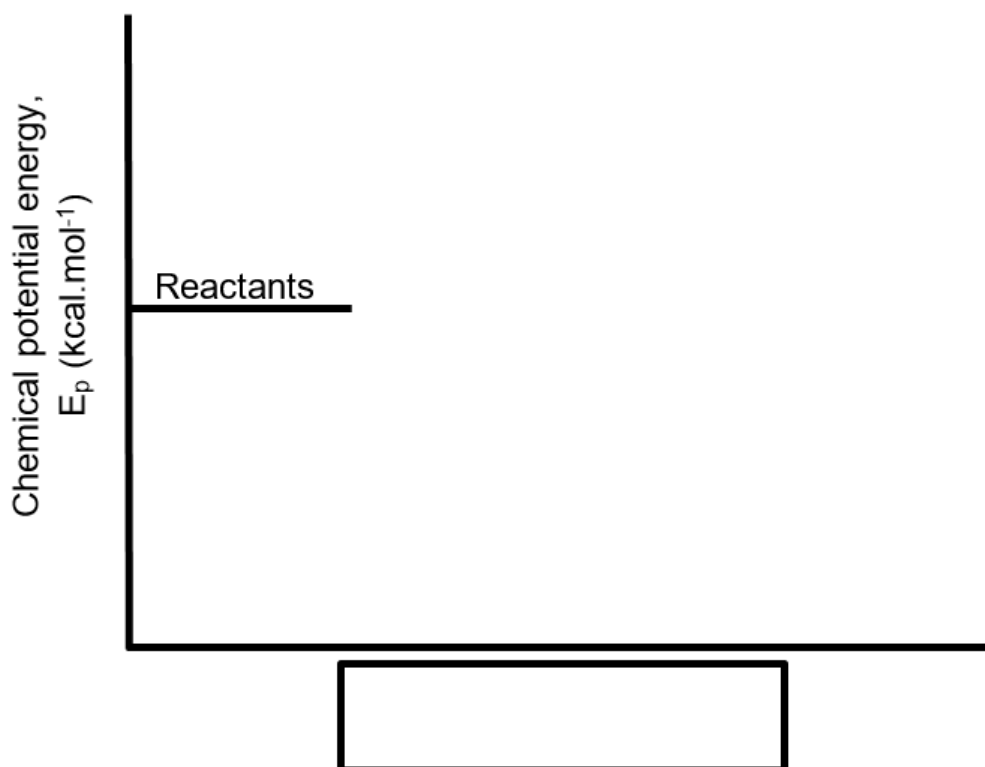
- 4.1 Define heat of reaction (ΔH). (2)

- 4.2 For reaction (i), the activation energy, E_a , is 8 kcal.mol^{-1} and the energy released when the bonds form, E_{out} , is $54,1 \text{ kcal.mol}^{-1}$.

- 4.2.1 Calculate ΔH for the reaction. (3)

- 4.2.2 Complete the energy-profile graph for the reaction on the axes provided. (2)

- 4.2.3 Write the relevant labels in the boxes for the x-axis. (1)



- 4.2.4 Indicate and label the two energy intervals E_a and ΔH on your graph. It is not necessary to show the numerical values. (2)

4.2.5 A catalyst is now added to the reaction mixture.

a) Name a suitable catalyst. _____ (1)

b) A learner incorrectly states:

"A catalyst lowers the activation energy of the reaction which results in more collisions taking place per second."

Explain why this statement is incorrect. (2)

c) Draw in a second graph to show the effect of the catalyst on the reaction. (2)

QUESTION 5

34 MARKS

(Adapted from the IEB 2022 Paper 2)

Nitrogen monoxide gas may be prepared by adding copper metal to dilute nitric acids. The balanced equation for the redox reaction is given below:



BLUE

Three different experiments were conducted in which various reactions conditions were changed. The data is tabulated below:

Experiment	Mass of Cu(g)	Volume of HNO ₃ (dm ³)	Concentration (mol.dm ⁻³)	State of Cu(s)
A	2,54	0,10	0,8	Small pellets
B	2,54	0,05	0,8	Large chunks
C	5,08	0,10	1,6	Small pellets

5.1 Using the data from EXPERIMENT A:

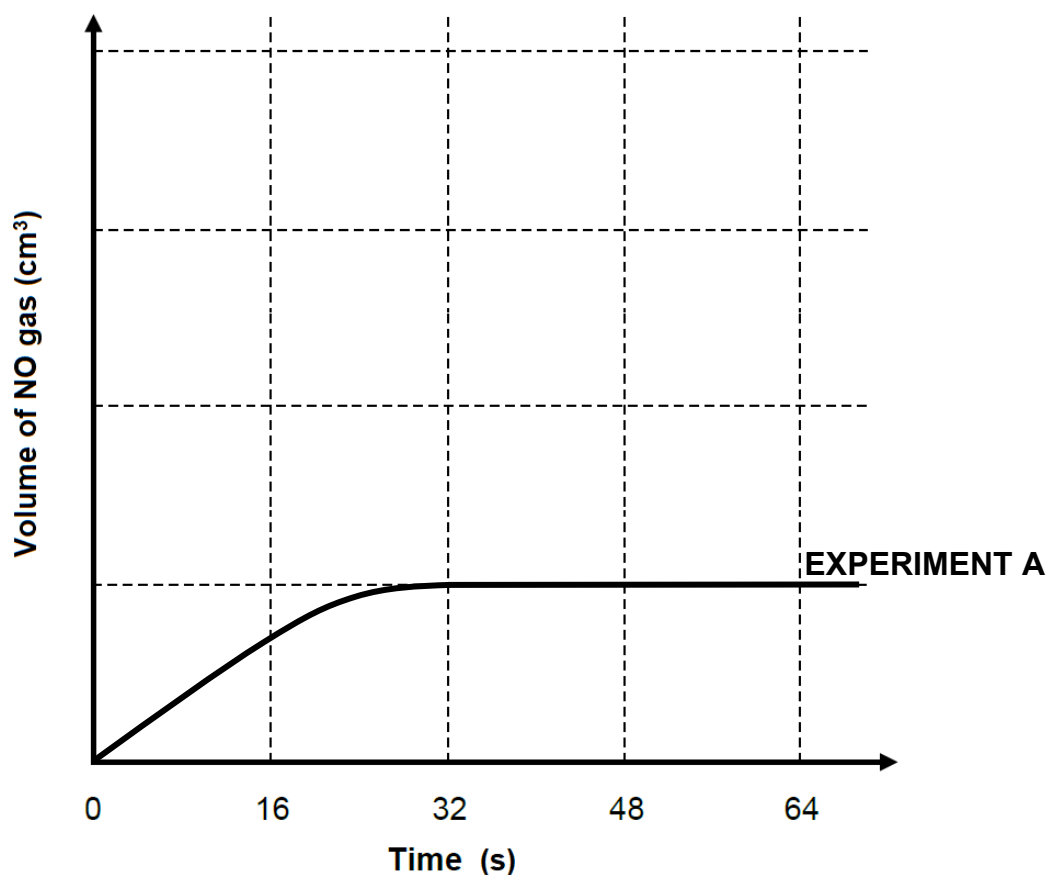
5.1.1 Calculate the number of moles of Cu. (3)

5.1.2 Calculate the number of moles of HNO_3 . (3)

5.1.3 Hence, show that copper is in excess. (3)

5.1.4 Assuming that the reaction proceeds to completion, calculate the volume of NO(g) that would be produced at STP. (4)

- 5.2 The NO gas was collected in a gas syringe. The volume of the NO gas produced in EXPERIMENT A is plotted against time, as shown in the graph below:

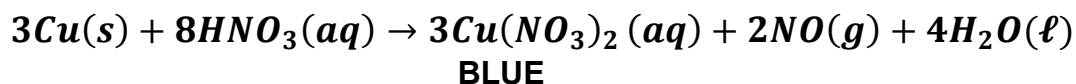


- 5.2.1 Calculate the average rate of the reaction for EXPERIMENT A (3)

- 5.2.2 Draw the curves that will be obtained for experiment B and C on the axes. **LABEL** the **two** new curves clearly. (4)

- 5.2.3 Explain how an increase in concentration affects the rate of a reaction in terms of Collision Theory. (4)

The reaction equation is re-written below:



5.3 Some alternative methods of measuring the rate of this reaction are:

- Measuring the mass loss of the reaction mixture vs. time.
- Shining light through the solution and measuring the % transmission of the light vs. time.

5.3.1 State whether the mass loss of the reaction mixture would INCREASE, DECREASE or STAY THE SAME as the reaction proceeds. Give a reason for your answer. (3)

5.3.2 State whether the % transmission of the light through the reaction mixture would INCREASE, DECREASE or STAY THE SAME as the reaction proceeds. Give a reason for your answer. (3)

5.4 The reaction equation is re-written below:



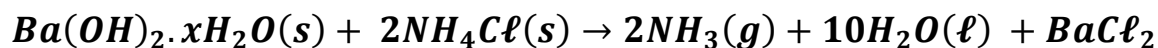
5.4.1 Define redox reaction. (1)

5.4.2 Write the oxidation half reaction for a copper atom. (2)

5.4.3 Is Cu the reducing agent in the reaction above?
Circle: YES or NO. (2)

QUESTION 6**28 MARKS**

Solid barium hydroxide and solid ammonium chloride react when mixed in a beaker. The reaction produces a liquid, and the temperature drops to -25°C .



6.1 Consider the heat of reaction.

6.1.1 Define *endothermic reaction*. (2)

Circle your answers for the following questions.

6.1.2 In the reaction above, more energy is absorbed than is released. Classify the reaction as ENDOTHERMIC or EXOTHERMIC. (1)

6.1.3 In order for the reaction above to proceed, some bonds must be broken and others must be formed. Is bond breaking an ENDOTHERMIC or EXOTHERMIC process? (1)

6.1.4 Would the products contain MORE or LESS potential energy than the reactants? (1)

6.1.5 Would the heat of reaction be POSITIVE or NEGATIVE? (2)

6.2 Consider the solid ammonium chloride.

6.2.1 Define *ionic bond*. (2)

6.2.2 The ammonium chloride compound has one cation and one anion. Draw a Lewis diagram of each ion. (4)

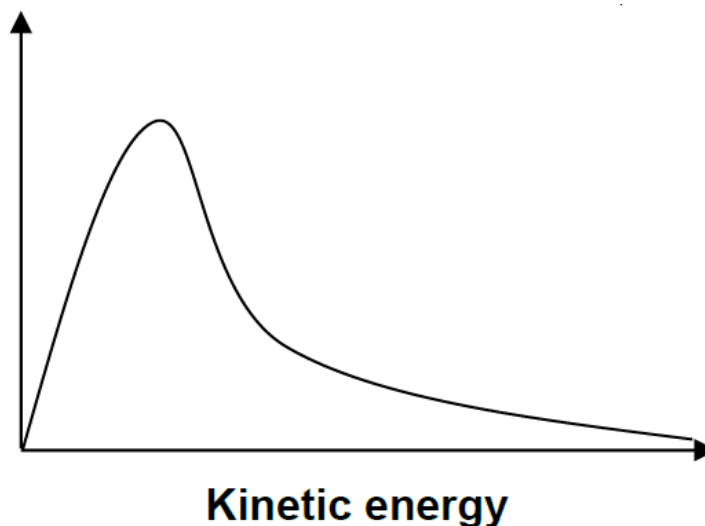
Cation	Anion

6.4 Consider rate of reaction.

6.4.1 Define rate of reaction.

(2)

6.4.2 The following graph shows the Maxwell-Boltzmann distribution curve for the reaction at room temperature (25 °C).



On the graph on the above, draw the curve that would be obtained at 50 °C.

Indicate the following on your graph:

- The Y-axis label.
- E_A , the activation energy.
- P, a shaded area representing the proportion of particles with sufficient kinetic energy to react.

(4)

QUESTION 7

23 MARKS

Hexane and 2,2-dimethylbutane are isomers.

7.1 Define structural isomer. (2)

7.2 Write down the general formula for the homologous series to which the compounds belongs. (2)

7.3 Given the molecular formula for both compounds. (2)

7.4 Draw the structural formulae for each of these substances. (5)

Hexane

2,2-dimethylbutane

7.5 Classify the compounds as SATURATED or UNSATURATED. Circle your answer. (1)

7.6 Which of hexane and 2,2-dimethylbutane has the stronger intermolecular forces? (1)

7.7 Explain your answer to 7.6. (3)

- 7.8 Draw the structural formula for the POSITIONAL ISOMER of compound 2,2-dimethylbutane. (3)

- 7.9 Hexane undergoes the following reaction:



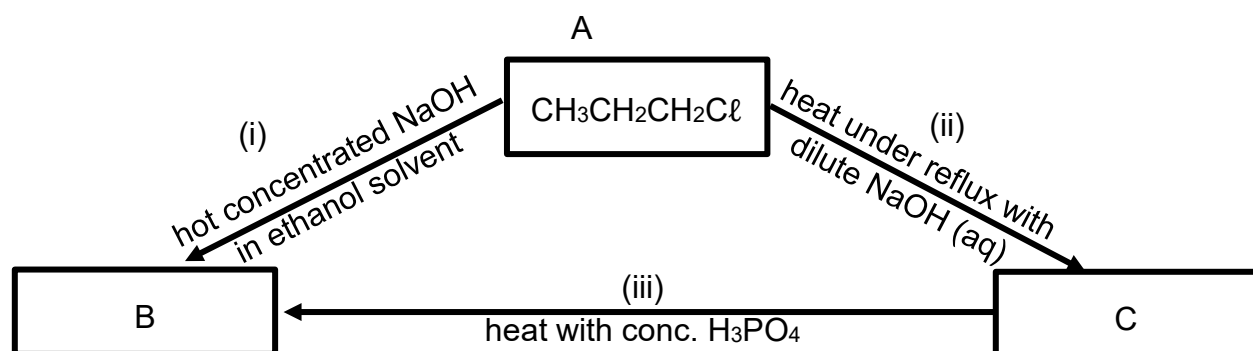
- 7.9.1 State the SPECIFIC reaction type. (2)

- 7.9.2 Give the IUPAC name of compound X. (2)

QUESTION 8

17 MARKS

Study the flowchart below in which B and C are the organic products of chemical reactions.

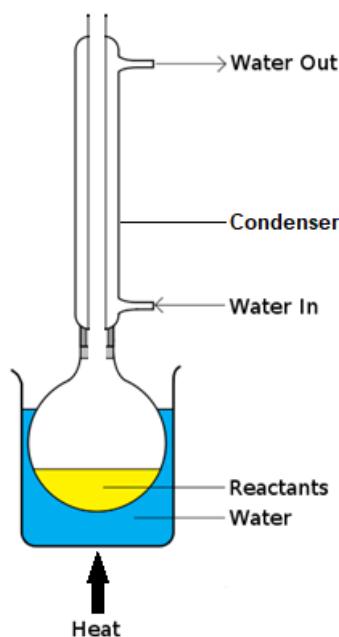


- 8.1 Write the IUPAC name for compound A. (2)

- 8.2 Define homologous series. (2)

- 8.3 Identify the homologous series to which compound A belongs. (1)

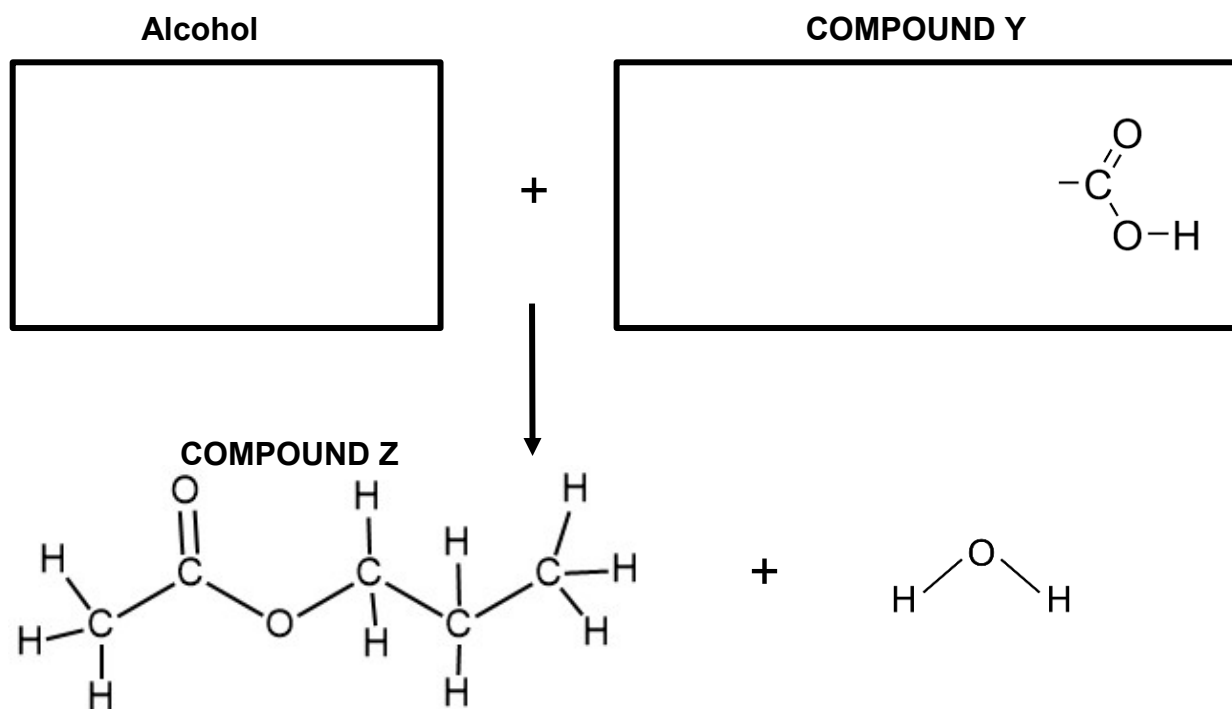
- 8.4 Write the condensed structural formula for:
- 8.4.1 compound **B**. _____ (2)
- 8.4.2 compound **C**. _____ (2)
- 8.5 For reaction (i), state the GENERAL reaction type. _____ (1)
- 8.6 For reaction (ii), state the SPECIFIC reaction type. _____ (2)
- 8.7 For reaction (iii), state the SPECIFIC reaction type. _____ (2)
- 8.8 The diagram illustrates 'heating under reflux'. In reaction (ii).



Why is it often necessary to carry out reactions involving organic compound under reflux? (2)

QUESTION 9**30 MARKS**

- 9.1 Consider the reaction between two molecules combining to form a single molecule, with the elimination of a water molecule.



- 9.1.1 State the GENERAL term used to describe the reaction above. (1)

-
- 9.1.2 Define *functional group*. (2)

-
- 9.1.3 Name the functional group of compound Y. (1)

-
- 9.1.4 Circle the functional group of COMPOUND Z (1)

- 9.1.5 Name the homologues series of compound Y. (1)

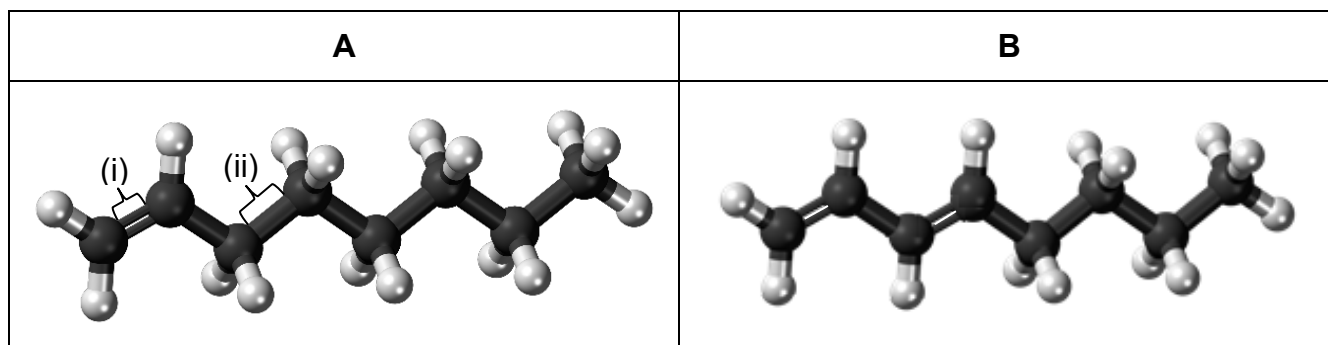
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- 9.1.6 Name the homologues series of compound Z. (1)

-
- 9.1.7 In the boxes, use structural formula:

- a) to draw the alcohol used in the reaction (2)
- b) complete compound Y (2)

9.1.8 Write a balanced chemical equation using chemical formula for the combustion reaction of the alcohol. (3)

9.2 Consider the compound A and B.



9.2.1 Give the IUPAC name of compound B. (3)

9.2.2 Consider the bonds (i) and (ii) in compound A. Account for the difference in length. (3)

9.2.3 Red-brown bromine water be used to distinguish between the two compounds.

a) Explain why red-brown bromine water can be used? (3)

b) Describe the method to distinguish the two compounds? (4)

[illegible]

[illegible]

[illegible]